



Purpose

The *Use Case Risk Classification Framework* scores a proposed agentic deployment by its inherent risk, independent of the organisation deploying it. It produces a single risk profile, Low, Medium, High, or Safety-critical, from eight dimensions. The profile is one of two inputs to the deploy decision; the other is organisational readiness, measured by the *Agentic AI Readiness Assessment*. The board's risk appetite is applied on top of both.

The framework is deliberately organisation-neutral. The same use case yields the same profile in any organisation; what differs is the appetite the board sets and the readiness the organisation holds. Appetite is not a dimension here. It is the tolerance line drawn across the result.

The eight dimensions

Decision authority	Does the agent recommend, or act?
Reversibility	Can the action be undone?
Data sensitivity	What class of data does it touch?
Blast radius	Who is affected if it fails?
Regulatory exposure	What regime governs the outcome?
Financial materiality	What is at stake, against a human sign-off analogy?
Human oversight	Where does the human sit relative to the loop?
Velocity	How fast does it act, and how wide is the intervention window?

The four risk profiles

Low	Advisory or fully reversible; low-sensitivity, narrow blast, no specific regulation.
Med	Bounded autonomy, reversible at cost; internal data, team-level blast, general regulation.
High	Autonomous within policy, hard to reverse; personal or financial data, BU-level blast, sectoral or EU AI Act exposure.
Crit	Unsupervised autonomy, irreversible; special-category data, enterprise or physical blast, safety regulation.

How to classify

Rate each dimension against the column that matches the proposed deployment. The overall profile is the highest single dimension, not the average. A use case that is Low on seven dimensions and Safety-critical on one is Safety-critical: the severity of an agent is set by its worst exposure, not its typical one.

This framework provides an evaluation methodology, not regulatory advice or an audit opinion.



The classification worksheet

Eight dimensions, four profiles. Rate each dimension by the column that matches the proposed deployment; the overall classification is the highest single dimension.

Dimension	Low	Medium	High	Safety-critical
Decision authority	Advisory; a human decides.	Acts within narrow pre-approved bounds; reviewed before effect.	Acts autonomously within policy; reviewed after the fact.	Acts autonomously; no human decision at run time.
Reversibility	Fully reversible at no cost.	Reversible at modest cost or delay.	Costly, slow, or only partly reversible.	Irreversible.
Data sensitivity	Public or non-sensitive.	Internal or commercially sensitive.	Personal (PII) or regulated financial.	Special-category personal, or safety-critical operational.
Blast radius	One task, one user.	One team or process.	A business unit or many customers.	Enterprise, public, or physical assets and people.
Regulatory exposure	No specific regulation.	General data-protection or consumer baseline.	Sectoral regulation; EU AI Act obligations.	EU AI Act high-risk boundary, or safety regulation.
Financial materiality	Below routine spend authority.	Within delegated budget.	Would need executive or CFO sign-off.	Board-level; could affect solvency or the accounts.
Human oversight	In the loop; approves each action.	On the loop; monitors, can intervene live.	Above the loop; sets policy, reviews periodically.	No effective oversight at decision time.
Velocity	Batch; hours or days to review.	Near-real-time; minutes.	Real-time; seconds.	Sub-second, high-frequency; no human-speed intervention.

Highest-dimension rule. The overall classification is the worst single dimension, not the average. One Safety-critical dimension makes the use case Safety-critical, however benign the rest. Reversibility, blast radius, and velocity together set how much the governance window can be trusted: an irreversible action at machine speed leaves no room to correct after the fact, which is why such a use case must be governed before it acts.



How classification connects to deployment

The risk profile is one leg of a three-way test. The second is the organisation's readiness, scored by the *Agentic AI Readiness Assessment*. The third is the board's risk appetite, set at the board as a fiduciary act. The CDO, with the Chief Risk Officer and legal counsel, operationalises that appetite into the *Minimum Maturity Deployment Thresholds*: for each profile, the minimum readiness required on the capabilities the use case depends upon. At the Deployment Gate, the CDO checks the profile against the readiness under those thresholds, and attests or halts.

The relationship is monotonic. A higher profile demands higher readiness, with the Enforcement aspect at least level with the Technology aspect in the dependent zones. A fast, irreversible, autonomous agent must be governed by constraints that fire before it acts, not by dashboards that notice afterwards.

Where the thresholds live

This framework classifies; it does not set thresholds. Minimum maturity thresholds are organisation-specific, regulator-specific, and appetite-specific, and they carry legal exposure. They are set internally by the CDO with the CRO and legal counsel, encoded as SHACL guardrails, and enforced at the gate. Publishing them would create false comfort and legal risk. This instrument provides the classification side of the conversation.

About the author

Frédéric Verhelst, PhD (Applied Physics, TU Delft). Twenty-five years across data, AI, and industrial operations. Twelve years governing autonomous decision-support systems where machine error is financially and physically catastrophic: ISAPP closed-loop reservoir management (Shell, TNO, TU Delft); BDI multi-agent systems and integrated operations centres at Epsis (NCS, Inpex Ichthys). Then, as external executive advisor, the digital strategy and value case for the Tyra Redevelopment Final Investment Decision at Mærsk Oil (joint-venture partners Shell and Chevron); then Head of Data Office at TotalEnergies EP Danmark, and CEO of the KGA. Most recently bringing agentic AI into safety-critical maritime operations. He partners with C-suites on CDO mandates that convert unquantifiable agentic AI exposure into a calculable, insurable damage envelope. Currently available for non-executive director (NED) and CDO appointments focused on corporate AI governance.

Canonical reference: *The Agentic AI Capability Stack™*, Version 3.1. Companion artefacts: *Agentic AI Readiness Assessment v1.1*, *CDO Mandate v1.2*, *Board Briefing v1.1*, *Vendor Coverage Diagnostic Q2 2026 v1.1*, *CDO JD v1.1*. All artefacts available at fredericverhelst.com/TOI-library. Licensed CC BY-ND 4.0 with attribution.

This framework provides an evaluation methodology. It is not regulatory advice and does not constitute an audit opinion. Risk appetite is set by the board; minimum maturity thresholds per risk profile are set by the organisation's Chief Data Officer with the Chief Risk Officer and legal counsel.